# Python and Git 101

Python Programming and Data Analysis · Day 1 · Dataminded Academy

dataminded

NICE TO MEET YOU

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Your trainer today. I like Rust, playing the saxophone and triathlon.

Platform engineer and Dataminded Academy lead



# AI is a multiplier of experience

your experience × AI = what you get

Zero experience gives AI nothing to multiply.

### Speak the language

Know the words a specialist uses, so you can ask for exactly that

### Understand the tradeoffs

There are several ways to do it, each with a cost. You choose

### Ask specific questions

A precise request gets code you can read and check

# This prompt will make sense by 15:00

```
Write a function `summarize_run(path: Path) -> dict` that reads a CSV
log with `csv.DictReader`, converts `pressure_bar` to float, skips and
counts rows that raise `ValueError`, and returns the mean and max. Use
a list comprehension where it fits. Standard library only, no classes.
Add a pytest test using `pytest.approx`.
```





# By 17:00 you can do six things on your own laptop

### Run a script

Set up Python and run a script from PowerShell

### Manage a project

A virtual environment per project, packages with uv

### Write a program

Functions, conditions, loops and imports, run from VS Code

### Read an error

Find the line a stack trace points to

### Automate files

Read a folder of files and write a summary

### Work with Git

Clone, commit on a branch, push and pull from GitHub

# Agenda

| | |
|---|---|
| 01 | 09:00 Setup check |
| 02 | 09:30 Python on your laptop, scripts and notebooks |
| 03 | 10:15 Language fundamentals, lunch at 12:30 |
| 04 | 13:15 Files, the standard library and exceptions |
| 05 | 14:20 Projects with uv |
| 06 | 15:05 Version control and Git |
| 07 | 16:50 Wrap-up |





# Check the tools, then clone the repository

### 1 · The tools answer

```
uv --version
uv python list --only-installed
code --version
git --version
```

### 2 · The course material is on your laptop

```
cd $HOME\Documents
git clone https://github.com/datamindedacademy/ac-python-and-git-101
cd ac-python-and-git-101
uv sync
code .
```



**Something fails?**

Do not debug alone. Call me over: setup problems are normal on managed laptops.





# Python on your laptop



# Python fills the gap between shell scripts and C

### Shell scripts

Quick to write. Painful once you need real logic, numbers or data structures

### Python

Reads like a script, with lists, dicts, functions and a large library built in

### C

Can do everything. Slow to write, and easy to get wrong



Photo: Daniel Stroud, CC BY-SA 4.0

Guido van Rossum started it in 1989 at CWI in Amsterdam, for system administration work that was too much for a shell script and not worth a C program. First release in 1991.

# Python is the language, CPython runs it

**Your .py file** → **Compiled to bytecode** → **Run by the virtual machine**

The **interpreter** is the program that does the last two steps: `python.exe` .
To view the bytecode of a script: `uv run python -m dis <script.py>`

| | CPython | PyPy | MicroPython |
|---|---|---|---|
| **What it is** | The reference implementation, written in C | Compiles hot loops to machine code while running | A small Python for microcontrollers |
| **When you meet it** | Always: it is what uv installs | Long pure-Python calculations | Sensors and embedded boards |





# Check which Python a command really runs

```
Get-Command python
```

```
CommandType  Name        Source
-----------  ----        ------
Application  python.exe  C:\Users\you\AppData\Local\Microsoft\WindowsApps\python.exe
```

**Watch out**

A `WindowsApps` path is the Microsoft Store shortcut, not a real Python. It opens the Store or does nothing useful.

**PATH** is the list of folders Windows searches, top to bottom, when you type a command.





# uv is one tool for Python, packages and running

1

## INSTALLS PYTHON

`uv python install 3.13` puts a real CPython in your user folder, no admin rights needed

2

## INSTALLS PACKAGES

`uv sync` reads the project file and creates `.venv`, a **virtual environment**: a folder with the packages this project needs

3

## RUNS YOUR CODE

`uv run python script.py` uses that Python and those packages, every time

> You did steps 1 and 2 in the setup check. Step 3 is how we run everything today. Creating your own project with uv comes this afternoon.





# Go to the repository, then uv run the script

```
# exercises/python/01-first-script/hello.py
print("Hello, test engineer!")
```

```
cd $HOME\Documents\ac-python-and-git-101
uv run python exercises/python/01-first-script/hello.py
```

```
Hello, test engineer!
```

> Paths are read from your **working folder**, the folder PowerShell is in. Check it with `pwd`.





# VS Code runs the same script in three steps

1

## OPEN THE FOLDER

File, Open Folder, choose `ac-python-and-git-101`

2

## SELECT THE INTERPRETER

Command Palette, Python: Select Interpreter, pick the one in `.venv` or the uv Python 3.13

3

## RUN

Play button at the top right, or the same `uv run python` command in the integrated terminal





# Scripts and notebooks



# A script is a text file that runs top to bottom

```
# bench_summary.py
pressure_bar = 7.5

power_kw = 37.5
print(f"Power: {power_kw} kW at {pressure_bar} bar")

print("Done")
```

```
Power: 37.5 kW at 7.5 bar
Done
```

# A notebook runs code cell by cell

```
In [1]   pressure_bar = 7.5
```

```
In [2]   pressure_bar * 14.5038
```

```
Out [2]  108.7785
```

```
In [3]   print(f"{pressure_bar} bar")
```

```
Out      7.5 bar
```

- A file with the extension `.ipynb`, opened in VS Code or Jupyter
- You run one **cell** at a time with `Shift+Enter`, in any order
- The **kernel** is the Python process behind it. It keeps your variables between cells
- Output, tables and plots are saved in the file, next to the code





# Scripts repeat, notebooks explore

| | Script (.py) | Notebook (.ipynb) |
|---|---|---|
| **Runs** | Start to finish, every time, on demand or scheduled | Cell by cell, in any order you click |
| **Great for** | Automation, reusable tools, anything others run | Exploring data, plotting, explaining results |
| **Weak at** | Slow feedback when exploring | Repeatability and testing |
| **In Git** | Clean text, easy to compare versions | JSON with embedded output, noisy differences |





# A notebook has hidden state

```
In [1]   limit_bar = 9.0
```

```
In [2]   limit_bar = 6.0
```

ran, then deleted

```
In [3]   print(limit_bar)
```

```
Out      6.0
```

not 9.0

- The **kernel** remembers every cell you ran, also the one you deleted. That memory is the **hidden state**
- Cell 3 prints a value that appears nowhere in the visible code
- A colleague runs the notebook top to bottom and gets a different answer: it is not **reproducible**
- Restart the kernel and run all cells before you trust a result





# Exercise 01: First script

`exercises/python/01-first-script`

**Words you can now use** script · notebook · cell · kernel · hidden state · reproducible · interpreter · CPython · terminal · PATH · working folder · uv run

# Language fundamentals

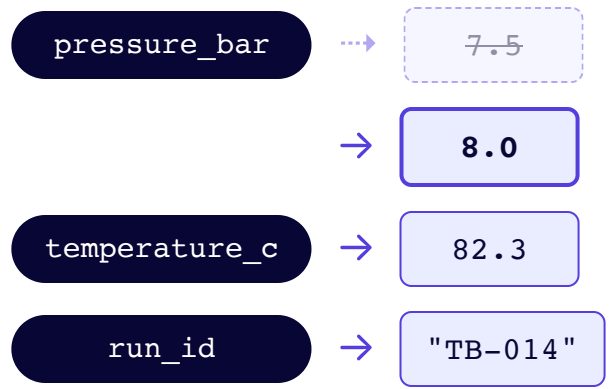


# Variables are names for values

```
pressure_bar = 7.5
temperature_c = 82.3
run_id = "TB-014"

pressure_bar = 8.0
print(pressure_bar)
print(temperature_c)
```

```
8.0
82.3
```




`=` is **assignment**: it gives a value a name.

**Question** After `pressure_bar = 8.0`, what happens to the bytes that held 7.5?

# Every value has a type

**Why** The type decides what an operation does: `7 + 5` is 12, `"7" + "5"` is "75".

```
run_id = "TB-014"        # str: text, a string
cycles = 12              # int: whole number
flow_l_s = 18.4          # float: decimal number
is_stable = True         # bool: True or False
operator = None          # None: no value (yet)

print(type(flow_l_s))
```

```
<class 'float'>
```





# Text that looks like a number is still text

```
text = "7.5"
pressure_bar = float(text)
print(pressure_bar + 1)        # 8.5

count = 3
print("Cycles: " + count)
```

```
TypeError: can only concatenate str (not "int") to str
```

> `float(text)` **converts** text to a number. Python never converts for you.

**Question** `"7.5" * 2` gives no error. What does it print?





# A traceback is a map to the error

```
Traceback (most recent call last):
  File "convert.py", line 2, in <module>
    pressure_bar = float(text)
ValueError: could not convert string to float: 'seven'
```

- Line 1 is only a header, it says: read on
- The middle shows the file, the line number and the code that failed
- The last line names the error type and what went wrong. Read it first.





# f-strings put values inside text

**Why** Every result you report is text with numbers in it: a log line, a file name, a plot title. An f-string sets the decimals and the unit in one place.

```
run_id = "TB-014"
pressure_bar = 7.5
print(f"Run {run_id}: {pressure_bar:.1f} bar")
print(f"{pressure_bar * 14.5038:.0f} psi")
print("TB" in run_id, run_id.lower())
print("12.5 kW".split())
```

```
Run TB-014: 7.5 bar
109 psi
True tb-014
['12.5', 'kW']
```

> A **method** is a function you call on a value with a dot: `run_id.lower()`





# A comparison gives a boolean: True or False

```
pressure_bar = 8.7
temperature_c = 82
is_alarm = pressure_bar > 8.5
print(is_alarm, type(is_alarm))
print(pressure_bar > 8.5 and temperature_c > 90)
print(pressure_bar > 8.5 or temperature_c > 90)
print(not is_alarm)
print(7.0 < pressure_bar < 9.0)
```

```
True <class 'bool'>
False
True
False
True
```

Compare with `>` `<` `>=` `<=` `==` `!=`. Combine with **and**, **or**, **not**. One `=` assigns, two `==` compare.





# A number has no unit, so put it in the name

**Why** Python stores 7.5, not 7.5 bar. Mix bar and psi and you get a wrong number, not an error.

**Written fast**

```
p = 7.5
t = 82
calc(p, t)
```

**Read easily**

```
pressure_bar = 7.5
temperature_c = 82
to_kelvin(temperature_c)
```

Put the unit in the name. Constants in capitals: `ALARM_LIMIT_BAR = 8.5`

# Exercise 02: Variables and types

`exercises/python/02-variables-and-types`

**Words you can now use** variable · assignment · type · string · float · f-string · convert · traceback · boolean



# A list keeps values in order

```
pressures_bar = [6.8, 7.1, 7.5, 7.4, 7.9]
print(pressures_bar[0])
print(len(pressures_bar))
pressures_bar.append(8.2)
print(max(pressures_bar))
print(sorted(pressures_bar, reverse=True))
```

```
6.8
5
8.2
[8.2, 7.9, 7.5, 7.4, 7.1, 6.8]
```





# Indexing starts at zero

| index | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| value | 6.8 | 7.1 | 7.5 | 7.4 | 7.9 |
| negative | -5 | -4 | -3 | -2 | -1 |

```
p = [6.8, 7.1, 7.5, 7.4, 7.9]
print(p[1:3], p[:2], p[-2:])
```

```
[7.1, 7.5] [6.8, 7.1] [7.4, 7.9]
```

A **slice** such as `p[1:3]` takes a part of the list. The end index is not included.

**Question** `p[10]` and `p[3:10]`: which one fails?





# A dict maps keys to values

```
reading = {"run": "TB-014", "pressure_bar": 7.5}
print(reading["pressure_bar"])
reading["flow_l_s"] = 18.4
print(reading.get("power_kw"))
print(reading)
```

```
7.5
None
{'run': 'TB-014', 'pressure_bar': 7.5, 'flow_l_s': 18.4}
```

Each entry is a **key** with its **value**: `"pressure_bar"` is a key, `7.5` is its value.

**Question** A dict with a million keys finds one as fast as a dict with ten. How?





# A tuple is a fixed group of values

```
position = (52.01, 4.36)
latitude, longitude = position
print(latitude)
position[0] = 0.0
```

```
52.01
TypeError: 'tuple' object does not support item assignment
```

- A **tuple** is written with round brackets and cannot be changed after it is made
- **Unpacking** gives each value its own name. A function that returns two values returns a tuple
- A **set** keeps only unique values: `set(["P1", "T1", "P1"])` is `{"P1", "T1"}`





# Four collections, picked by how you find a value

| | list | tuple | dict | set |
|---|---|---|---|---|
| **Find by** | Position, in order | Position, in order | Key | Is it in there? |
| **Change it?** | Yes | No | Yes | Yes, no duplicates |
| **Fits** | A series of the same kind of value | A fixed group that belongs together | One record with named fields | Unique values, membership |
| **Example** | `pressures[2]` | `lat, lon = pos` | `reading["run"]` | `"P1" in ids` |

> Copilot will pick one for you unless you say which you want.





# Exercise 03: Collections

`exercises/python/03-collections`

**Words you can now use** list · tuple · index · slice · dict · key · value · set · method



# if runs the first branch that is true

```
pressure_bar = 9.4
if pressure_bar > 9.0:
    print("ALARM: over limit")
elif pressure_bar < 5.0:
    print("Warning: low pressure")
else:
    print("OK")
```

```
ALARM: over limit
```

`pressure_bar > 9.0`
true: run this branch, skip the rest

`pressure_bar < 5.0`
not checked

`else`
not checked

Checked top to bottom, the first true one wins

The **condition** uses a **comparison operator**: `>` `<` `>=` `==` . **Indentation** of 4 spaces marks the block.





# `for` runs the block once per element

```
pressures_bar = [6.8, 7.1, 9.4]
for p in pressures_bar:
    print(f"{p:.1f} bar")

for i in range(3):
    print(i)
```

```
6.8 bar
7.1 bar
9.4 bar
0
1
2
```

To **iterate** is to go through a collection one element at a time. With the position: `for n, p in`





# Most scripts initialise, loop, test, collect

```
pressures_bar = [6.8, 7.1, 9.4, 7.4, 9.1]
high = []
total = 0.0

for p in pressures_bar:
    total += p
    if p > 9.0:
        high.append(p)

print(high, round(total / len(pressures_bar), 2))
```

```
[9.4, 9.1] 7.96
```

1. **INITIALISE**
   An empty list and a zero total
2. **LOOP**
   Take each value in turn
3. **TEST**
   Decide with an `if`
4. **COLLECT**
   Report at the end





# A list comprehension is that loop on one line

**The loop**

```
high = []
for p in pressures_bar:
    if p > 9.0:
        high.append(p)
```

**The comprehension**

```
high = [p for p in pressures_bar if p > 9.0]
```

```
# transform every value
psi = [p * 14.5038 for p in pressures_bar]

# the same idea without the brackets
total_high = sum(p for p in pressures_bar if p > 9.0)
```

> One `for`, one `if`. Anything more, write the loop.





# Exercise 04: Control flow

`exercises/python/04-control-flow`

**Words you can now use** condition · branch · loop · iterate · block · indentation · comprehension



# A function names a piece of logic

```
def to_psi(pressure_bar):
    return round(pressure_bar * 14.5038, 1)

print(to_psi(7.5))
print(to_psi(10))
```

```
108.8
145.0
```



# `return` passes a value on, `print` only shows it

```
def double_and_print(x):
    print(x * 2)

result = double_and_print(4)
print(result)
```

```
8
None
```

A function without `return` gives back `None`. Use `print` to show, `return` to pass on.

# A script is small functions plus a main

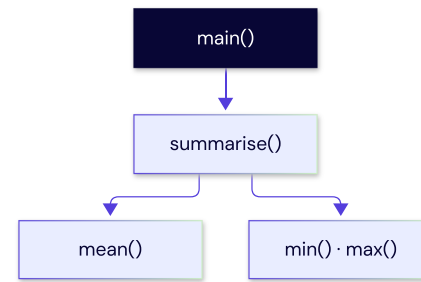
**Why** Small functions can be tested and reused one by one. `main` shows the order of the work in a few lines.

```
def mean(values):
    """Return the arithmetic mean of a list of numbers."""
    return sum(values) / len(values)

def summarise(values):
    return {"min": min(values), "max": max(values), "mean": mean(values)}

def main():
    pressures_bar = [6.8, 7.1, 9.4, 7.4]
    print(summarise(pressures_bar))

if __name__ == "__main__":
    main()
```




> The text in triple quotes is a **docstring**: it says what the function does.

# Type hints tell you how to call a function

```python
def to_psi(pressure_bar: float, decimals: int = 1) -> float:
    return round(pressure_bar * 14.5038, decimals)
```

### pressure_bar: float

The parameter and the type it expects

### decimals: int = 1

A type and a default value, so you may leave it out

### -> float

The type of the value that comes back

> Hints are documentation that the editor and tools can check. Python does not enforce them when the script runs.

**Question** The hint says `float`. What happens with `to_psi("7.5")`?

# import gives you ready-made tools like math

```
import math
from math import sqrt

diameter_m = 0.05
area_m2 = math.pi * (diameter_m / 2) ** 2
flow_m3_s = 18.4 / 1000
print(round(flow_m3_s / area_m2, 2))

print(math.isclose(0.1 + 0.2, 0.3), sqrt(16))
```

```
9.37
True 4.0
```





# Move code outward only when you need to

| | Use it when | The cost |
|---|---|---|
| **Lines in the script** | You do it once, in a few lines | Copy and paste when you need it again |
| **A function** | You repeat it, or it deserves a name | A little structure to read through |
| **A module** | Several scripts need the same functions | One more file, and an `import` |

> Copilot will pick one for you unless you say which you want.





# A test says what correct means

```python
import pytest
import conversions

def test_bar_to_psi():
    assert conversions.bar_to_psi(1) == pytest.approx(14.5038)
```

Red: run, read the failure → Green: make it pass → Test first: write the next test

This loop is **test-driven development**.

`assert` states an **assertion**: a claim that must be true, or the test fails.





# Exercise 05: Functions

`exercises/python/05-functions`

**Words you can now use** function · argument · return value · type hint · test · assertion · test-driven development · import · module



# Files and the standard library



# pathlib builds paths and finds files

```python
from pathlib import Path

data_dir = Path("data")
log_file = data_dir / "run_014.csv"

print(log_file.name, log_file.suffix, log_file.stem)
print(log_file.exists())

for f in sorted(data_dir.glob("*.csv")):
    print(f)
```

```
run_014.csv .csv run_014
True
data\run_014.csv
data\run_015.csv
```





# with open reads a file and closes it again

```
from pathlib import Path

with open("data/run_014.csv", encoding="utf-8") as f:
    for line in f:
        print(line.strip())

text = Path("data/run_014.csv").read_text(encoding="utf-8")
```

```
t_s,pressure_bar,temperature_c
0,6.8,71.2
10,7.1,74.5
20,7.5,78.9
```

`f` is the **file handle**. State the **encoding**, or Windows picks its own.





# Mode `w` overwrites, mode `a` appends

```
lines = ["run,max_pressure_bar", "TB-014,7.9"]
with open("summary.txt", "w", encoding="utf-8") as f:
    for line in lines:
        f.write(line + "\n")

with open("summary.txt", "a", encoding="utf-8") as f:
    f.write("TB-015,8.2\n")
```



Mode `"w"` replaces an existing file without asking. `"a"` adds at the end, `"r"` only reads.

# csv turns each row into a dict of strings

```
t_s,pressure_bar,temperature_c
0,6.8,71.2
10,7.1,74.5
20,7.5,78.9
```

```
import csv
with open("data/run_014.csv", newline="", encoding="utf-8") as f:
    rows = list(csv.DictReader(f))   # the header row gives the keys
print(rows[0])
print(float(rows[1]["pressure_bar"]) + 1)
```

```
{'t_s': '0', 'pressure_bar': '6.8', 'temperature_c': '71.2'}
8.1
```

**Question** The log has ten million rows. What does `list(...)` on line 3 do?





# json stores settings and nested data

**Why** CSV holds one flat table. Settings, limits and nested results need a format with names and levels.

```
import json
config = {"limit_bar": 9.0, "channels": ["pressure", "flow"], "operator": None}
text = json.dumps(config)
print(text)
back = json.loads(text)
print(back["channels"][0], back["operator"])
```

```
{"limit_bar": 9.0, "channels": ["pressure", "flow"], "operator": null}
pressure None
```





# Exercise 06: Standard library

`exercises/python/06-standard-library`

**Words you can now use** standard library · path · glob · file handle · encoding · CSV · JSON



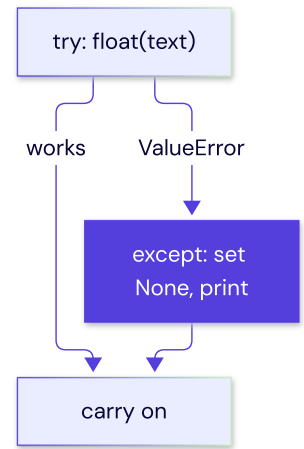
# Exceptions



# try / except handles an error you expect

```
text = "seven"
try:
    pressure_bar = float(text)
except ValueError:
    pressure_bar = None
    print(f"Skipping bad value: {text!r}")
print(pressure_bar)
```

```
Skipping bad value: 'seven'
None
```




`float("seven")` **raises a** `ValueError`**.** `except` **catches it.**

**Question** What changes if you write `except:` without `ValueError`?

# No plan for an error? Let it bubble up

**Swallowed**

```
try:
    mean_bar = summarise(path)
except Exception:
    pass
```

The script carries on. The report is wrong or empty, and nothing tells you why.

**Bubbles up**

```
mean_bar = summarise(path)
```

The error travels up through every function that called this one. Nobody catches it, so the script stops with a traceback: file, line and reason.

- Catch an error only where you can do something with it: skip the row and count it, use a default, ask again
- Need to add context? Catch it, then `raise` again with a better message
- A script that stops with a traceback is better than a script that finishes with a wrong number





# Exercise 07: Files and exceptions

`exercises/python/07-files-and-exceptions`

**Words you can now use** exception · raise · catch · bubble up



# Projects with uv



# Python packaging has a messy history



2007

**virtualenv**

a separate set of packages per project



2008

**pip**

installs packages, and nothing else



2012

**conda**

packages and Python itself, in its own world



2017

**pipenv**

pip and virtualenv in one, with a lock file



2018

**Poetry**

one project file, a lock file, publishing



2024

**uv**

all of it in one fast tool

- Each tool fixed one piece: packages, environments, Python versions, locking
- Also in this crowd: `pyenv`, `pip-tools`, Hatch, PDM, Rye
- What they agreed on in the end is one project file: `pyproject.toml`
- You will meet all of these in tutorials, old projects and Copilot answers

# uv and pip: the same job, fewer steps

| | pip workflow | uv workflow |
|---|---|---|
| **Environment** | `python -m venv .venv` | `uv init`, then automatic |
| **Activate** | `.venv\Scripts\Activate.ps1` | Not needed: `uv run` |
| **Install** | `pip install matplotlib` | `uv add matplotlib` |
| **Declare** | `requirements.txt`, by hand | `pyproject.toml`, updated by uv |
| **Pin versions** | `pip freeze`, if you remember | `uv.lock`, automatic |





# Two commands give a project its own packages

```
uv init bench-tools
cd bench-tools
uv add matplotlib
uv run python main.py
```

```
Creating virtual environment at: .venv
Resolved 12 packages in 330ms
Installed 11 packages in 68ms
 + matplotlib==3.11.2
 + numpy==2.5.4
 ...
```

> numpy came along without you asking, because matplotlib needs it: a **transitive dependency**.





# pyproject.toml declares, uv.lock pins

```
[project]
name = "bench-tools"
requires-python = ">=3.13"
dependencies = [
    "matplotlib>=3.11.2",
]
```

**Commit**

`pyproject.toml`, `uv.lock`, `.python-version` and your `.py` files

**Do not commit**

`.venv/`, big and machine specific

**Question** The script ran in March and fails in November. Nobody touched the code. What changed?





# One script, no project: the packages go in the file

```python
# /// script
# requires-python = ">=3.13"
# dependencies = [
#     "matplotlib>=3.11.2",
# ]
# ///
import matplotlib.pyplot as plt

plt.plot([7.1, 7.4, 7.6])
plt.savefig("pressure.png")
```

```
uv add --script plot.py matplotlib
uv run plot.py
```

- The comment block at the top is **inline script metadata**, a Python standard (PEP 723)
- `uv add --script` writes it for you
- `uv run` reads it, builds a temporary environment and runs the script

> One file to mail to a colleague: use this. Several files, tests or a lock file: make a project.





# Exercise 08: A uv project

`exercises/python/08-uv-project`

**Words you can now use** virtual environment · package · dependency · lockfile · pin · sync



# Python goes further, but not today

- **Styles**: procedural (today), object-oriented with classes and dataclasses, functional
- **Language**: decorators, generators, context managers
- **Debugging**: the VS Code debugger, with breakpoints and stepping, instead of `print`
- **Scripts that grow**: `logging` instead of `print`, arguments with `argparse`

- **Performance**: measure first (`timeit`, a profiler), then NumPy
- **Concurrency**: threads, processes, `asyncio`
- **Extending**: C, or Rust through PyO3
- **Packaging**: your own code, installable by colleagues





# Version control



# This prompt will make sense by 17:00

```
Fetch and rebase `feature/max-temperature` onto `origin/main`. If a commit
conflicts, stop and show me both sides. Then stage only `analyse_run.py`
and commit it as a conventional commit, type `fix`, scope `analysis`. Do
not commit `output/`. Push with `--force-with-lease` and draft the pull
request description from `git diff origin/main`.
```





# final_v3 is version control by hand

```
\\fileserver\compressors\testbench\
    analyse_run.py
    analyse_run_final.py
    analyse_run_final_v3_USE_THIS_ONE.py
```

> Git answers these four questions without copies of the file.

### History

Which version made last week's report?

### Who and why

Who changed the conversion factor?

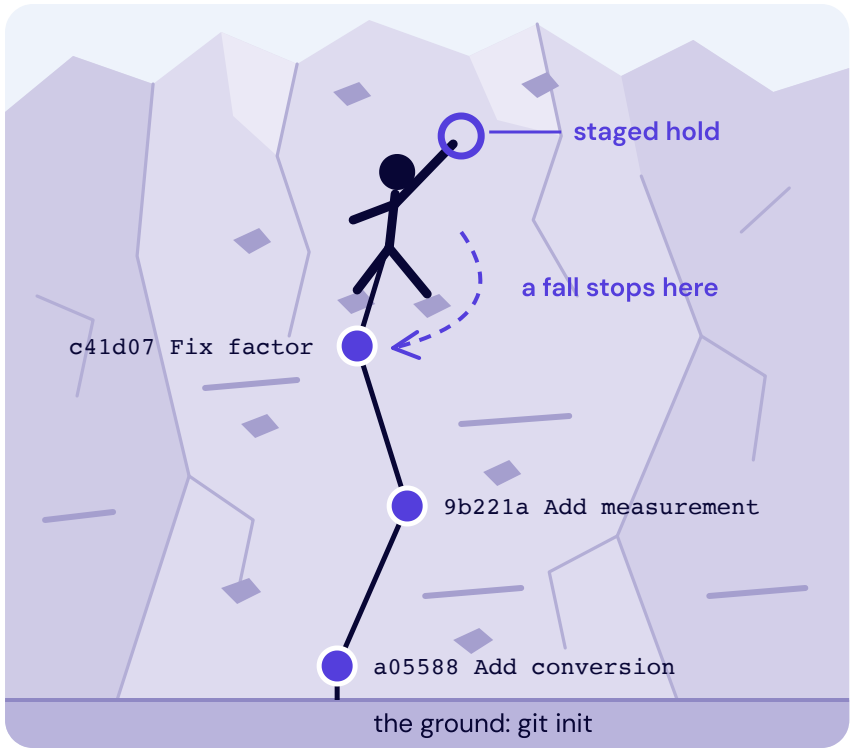
### Undo

How do I get back to what worked?

### Collaborate

Two people edit, whose version wins?

# Stage a hold, then hammer in an anchor



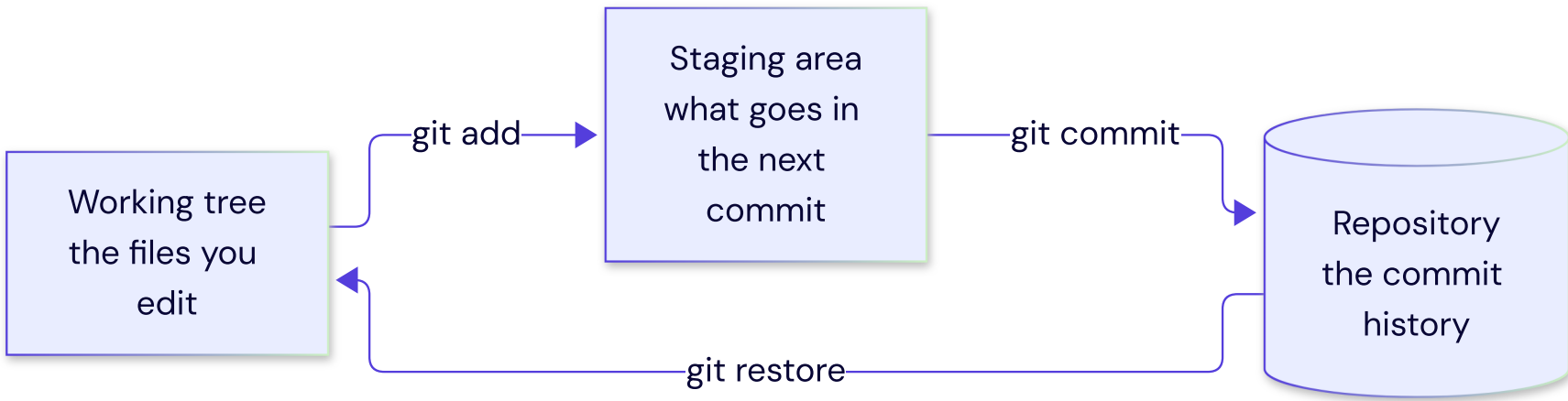

**Working tree**
You try moves above the last anchor. Nothing is saved yet.

**Stage** `git add`
You pick the holds for the next move.

**Commit** `git commit`
You hammer in an anchor and clip the rope. A fall takes you back to it, not to the ground.

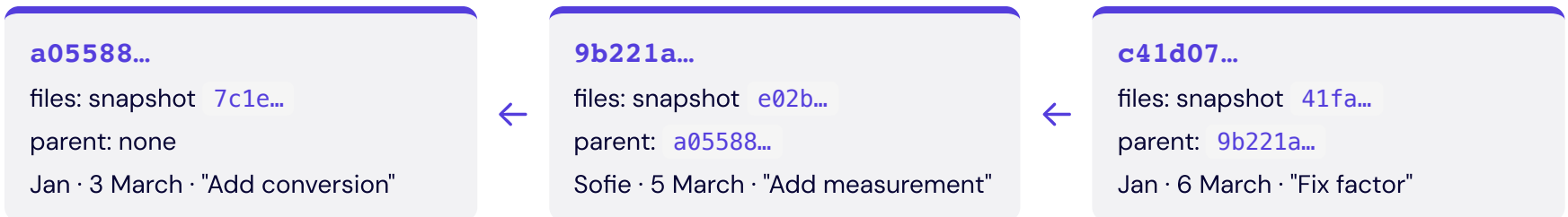
**History**
The line of anchors below you.

# A change moves through three areas




- You choose what goes into a commit by **staging** it first (the staging area is also called the **index**)
- `git status` tells you which area each change is in

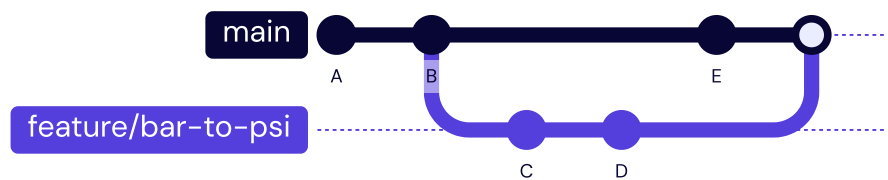
# A hash is the name Git gives to content



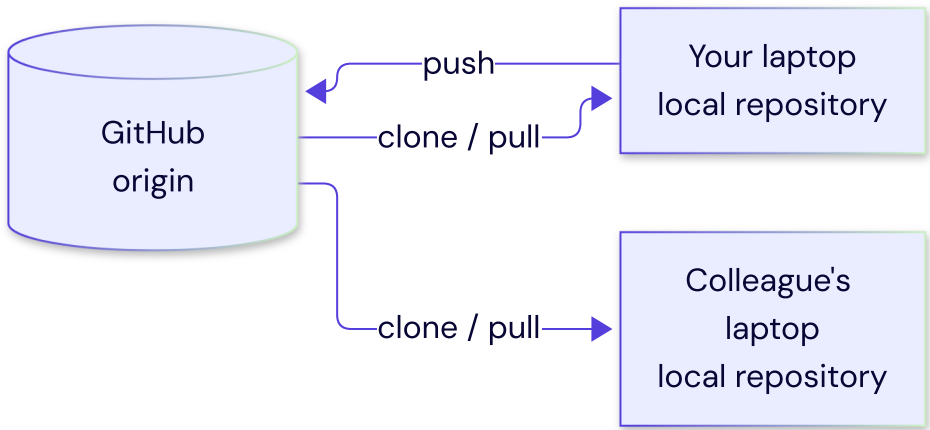

- Git stores every file and every commit under the **hash of its content**: 40 characters, calculated with SHA-1
- Same content, same hash. Change one character and the hash changes
- A commit contains its parent's hash, so changing an old commit changes every hash after it
- The first 7 characters are enough to point at a commit: `git show c41d07a`

# A branch is a movable label on a commit

- **Repository**: a project folder plus its full history
- **Commit**: a saved snapshot with a message, author and date
- **Branch**: a movable label on a commit
- Work on a branch does not touch `main` until you merge it



# GitHub hosts the shared remote




- **Git** is the free tool on your laptop, **GitHub** hosts the shared copy
- A **remote** is that shared copy, by convention called `origin`

**Words you can now use** repository · commit · branch · remote · working tree · staging area

# Git on the command line



# Use Git from the command line or VS Code

**VS Code**

Easier to use.

- Changed files, diffs and conflicts side by side
- Stage, commit and push with a click
- Next to the code you are editing

**Command line**

A skill that carries over.

- Works the same on a server, in a pipeline, on any laptop
- Every feature, and every error message and tutorial uses it
- Shows what Git does, step by step

An agent drives Git through commands. You can only judge what it did, or is about to do, if you can read them. So today: the command line first.

# Set your identity once, then init a repository

```
git config --global user.name "Your Name"
git config --global user.email "you@atlascopco.com"
mkdir testbench
cd testbench
git init -b main
git status
```

- Identity: once per laptop, same email as your GitHub account
- `git status` never changes anything: run it all the time

> A new file is **untracked**: Git sees it, but does not follow it until you `git add` it.





# Stage what belongs together, then commit

```
git add conversions.py
git commit -m "feat(conversions): add psi to bar conversion"
```

`type(scope): description` is a **conventional commit**. The scope is optional.

| Type | Use it for | Example |
|---|---|---|
| `feat` | new behaviour | `feat: report max temperature` |
| `fix` | a bug fix | `fix(conversions): correct psi to bar factor` |
| `docs` `test` `refactor` `chore` | no change in behaviour | `docs: explain the csv columns` |

- Description: short, imperative, one logical change. "and" in it means two commits
- `git add .` stages everything: check `git status` first





# log and diff show what changed

```
git log --oneline
git log --oneline --graph --all
```

```
9b221ad chore(data): add first measurement
a055887 feat(conversions): add psi to bar conversion
```

```
git diff            # edits not yet staged
git diff --staged   # what the next commit will contain
```

- `--graph --all` draws every branch as lines: you will read it in the branches exercise
- If the output is long, Git opens a pager: **q** leaves it

# Exercise time: first commit

`exercises/git/01-first-commit`

About 20 minutes: `uv run python exercises/git/setup_exercise.py 1`



# show and restore bring back old versions

```
git show a055887                    # one commit: message and diff
git show HEAD~1:conversions.py      # a file as it was one commit ago
git restore --source=HEAD~1 conversions.py
git restore conversions.py          # discard edits since last commit
```

- `HEAD` is the latest commit, `HEAD~1` the one before it
- `git restore <file>` throws your edits away: there is no undo for that





# Extra exercise: history

`exercises/git/02-history`

When you are ahead, or after class: `uv run python exercises/git/setup_exercise.py 2`



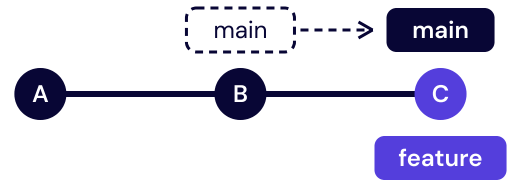
# Work on a branch, then merge it into main

```
git switch -c feature/bar-to-psi
git add conversions.py
git commit -m "Add bar to psi conversion"
git switch main
git merge feature/bar-to-psi
```

```
Updating 9b221ad..1cd9279
Fast-forward
```

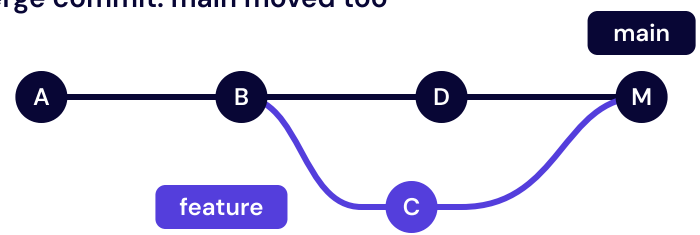
- `git branch` lists branches
- `git branch -d <name>` deletes one

**Fast-forward: main did not move**



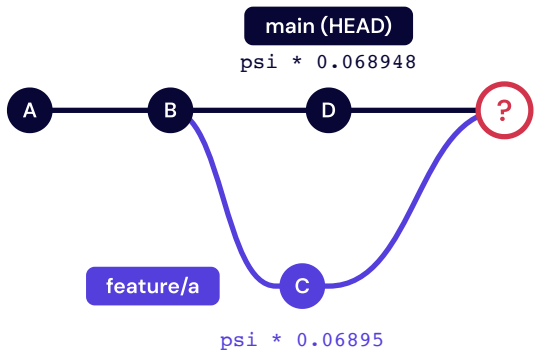

the label moves, no new commit

**Merge commit: main moved too**




M has two parents: D and C

# A conflict means Git needs you to decide




Both changed the same line after B.

```
CONFLICT (content): Merge conflict in conversions.py

def psi_to_bar(psi):
<<<<<<< HEAD
    return psi * 0.068948
=======
    return psi * 0.06895
>>>>>>> feature/a
```

**Above** `=======` is your branch. **Below** is the incoming branch.

- Nothing is lost. Edit the file, delete the marker lines, then `git add` and `git commit`
- Changed your mind? `git merge --abort`
- **Tip:** agents are excellent at solving merge conflicts. Read the result with `git diff` before you commit

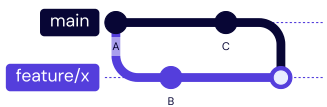
**Question** Two people change different lines of the same file. Is that a conflict?





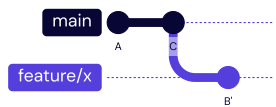
# Rebase replays your commits on top of main

**git merge main**




Keeps both lines, adds a merge commit.

**git rebase main**




Replays your commits after the latest `main`: a straight line, new hashes.

```
git switch feature/x; git rebase main
git add <file>; git rebase --continue   # after fixing a conflict
git rebase --abort                      # give up
```

> Rebase before you push. Rebasing rewrites commits, so a branch you already pushed needs `git push --force-with-lease` afterwards, and colleagues who pulled the old commits get duplicates.

# Exercise time: branches

`exercises/git/03-branches`

About 20 minutes: `uv run python exercises/git/setup_exercise.py 3`

**Words you can now use** stage · commit message · diff · HEAD · merge · fast-forward · conflict · rebase



# Remotes and GitHub



# Clone copies the repository and its history

```
git clone https://github.com/<organisation>/<training-repo>.git
cd <training-repo>
git remote -v
```

```
origin  https://github.com/<organisation>/<training-repo>.git (fetch)
```

- The first time, a browser window asks you to sign in (Git Credential Manager)
- `clone` creates the remote `origin`; the URL is under the green **Code** button on GitHub





# Push uploads commits, pull downloads them

```
git push -u origin feature/bar-to-psi   # first push of a new branch
git push                                 # later pushes
git pull                                 # fetch and merge new commits
git fetch                                # only download, change nothing
```

- `-u` sets the upstream: it links your local branch to the remote one, needed only once
- `origin/main` is a **remote-tracking branch**: your laptop's last known copy of `main` on the server, updated by `fetch` and `pull`
- `pull` is `fetch` plus merge: commit or restore your edits first

**Question** Your laptop is stolen on the train. Which work is gone?

# Exercise time: GitHub remote

`exercises/git/04-github-remote`

About 15 minutes: clone, branch, commit, push



# A pull request lets a colleague review first




### Propose

A pull request (PR) proposes merging your branch into `main`. Colleagues read the diff first.

### Merge

Click **Merge pull request** (squash keeps `main` readable), then **Delete branch**.

### Update your laptop

`git switch main`, then `git pull`

A **branch protection rule** on `main` can require a pull request with an approval before it can be merged.

# Exercise time: pull request

`exercises/git/05-pull-request`

About 20 minutes: PR into main, review, merge, pull



# .gitignore keeps files out of history

```
# .gitignore
.venv/
__pycache__/
data/raw/
```

- One pattern per line; commit the file itself so everyone shares it
- Never commit passwords, tokens or large raw data
- Day 2 uses it for data files and notebooks

> A **tracked** file is one Git already follows. `.gitignore` only works on files that are not tracked yet: they become **ignored**.





# Demo: Git in VS Code

Optional practice: `exercises/git/06-git-in-vscode`

In the **Source Control view**: + stages, Commit commits, **Publish Branch** is the first push, **Sync Changes** pulls and then pushes.

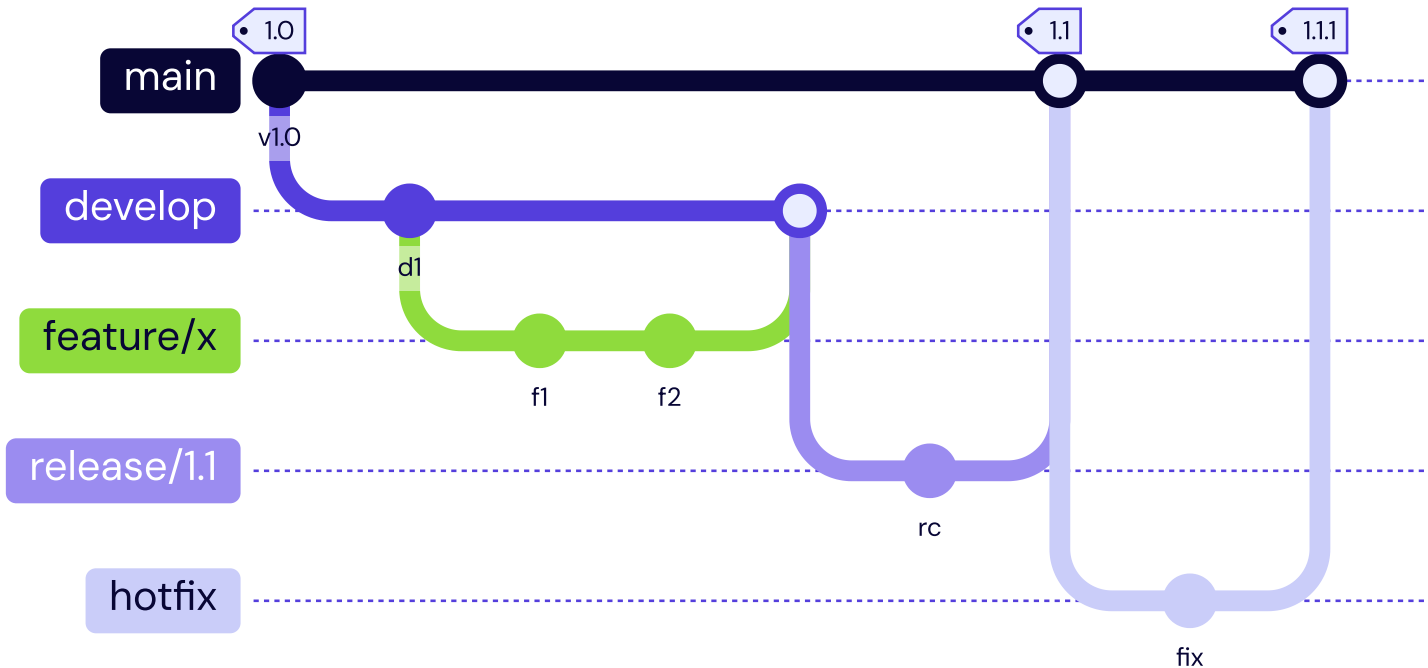
**Words you can now use** clone · origin · push · pull · fetch · upstream · origin/main · pull request · review



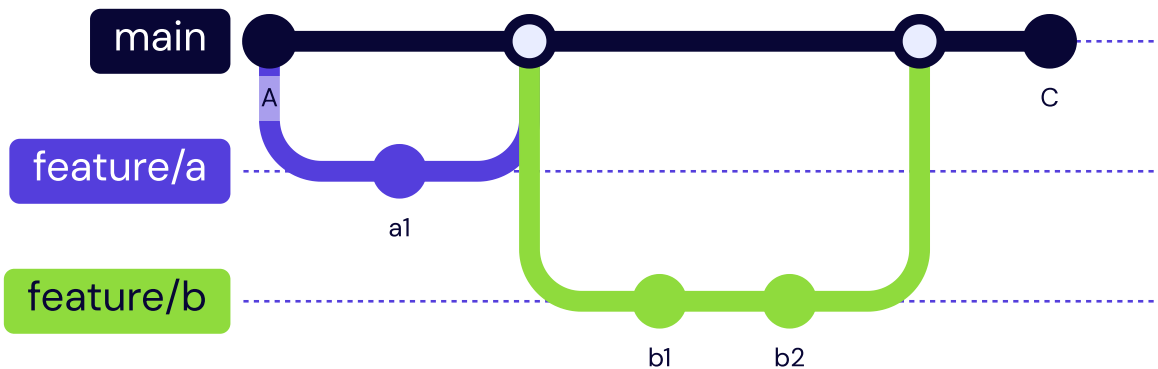
# Branching models



# GitFlow keeps a branch per release stage



# Trunk-based: one main, short branches




- One long-lived branch, `main`; short branches merged through a pull request

# Use trunk-based for scripts and analyses

| | GitFlow | Trunk-based |
|---|---|---|
| **Fits** | Versioned products, several releases in support | Continuous delivery, scripts, analyses, small teams |
| **Cost** | Many branches, large merges, more conflicts | Discipline: small changes, review, tests |

- **Use trunk-based**: short-lived `feature/...` branches and PRs; day 3 adds a branch protection rule on `main`
- Exists, not today: `git cherry-pick`, `git stash`

**Words you can now use** trunk-based · GitFlow · feature branch · short-lived · branch protection rule





# You can now do all six things

### Run a script

From PowerShell and VS Code

### Manage a project

A .venv per project with uv

### Write a program

Functions, loops, imports

### Read an error

Find the line in a stack trace

### Automate files

Read files, write a summary

### Work with Git

Branch, commit, push, PR, pull

Today: the **vocabulary** and the tradeoffs. Day 3: you use them to direct and review Copilot and coding agents.

# Read these two prompts again

## Git

```
Fetch and rebase `feature/max-temperature` onto `origin/main`. If a commit conflicts, stop and show me both sides. Then stage only `analyse_run.py` and commit it as a conventional commit, type `fix`, scope `analysis`. Do not commit `output/`. Push with `--force-with-lease` and draft the pull request description from `git diff origin/main`.
```

## Python

```
Write a function `summarize_run(path: Path) -> dict` that reads a CSV log with `csv.DictReader`, converts `pressure_bar` to float, skips and counts rows that raise `ValueError`, and returns the mean and max. Use a list comprehension where it fits. Standard library only, no classes. Add a pytest test using `pytest.approx`.
```





# Eight commands cover daily Git

## Local

```
git status
git add <file>
git commit -m "message"
git log --oneline
git diff
git restore <file>
git switch -c <branch>
git merge <branch>
```

## Remote

```
git clone <url>
git switch main
git pull
git switch -c feature/x
git push -u origin feature/x
# open the PR, merge it, repeat
```

# Before day 2: one full cycle with a PR

## Before day 2

- A working uv environment
- Redo the fundamentals you found hard
- One full clone, commit, push, pull cycle with a PR

## Read more

- Pro Git book ( `git-scm.com/book` )
- Learn Git Branching ( `learngitbranching.js.org` )
- Python tutorial ( `docs.python.org/3/tutorial` )
- uv docs ( `docs.astral.sh/uv` )

## Next days

- Day 2: pandas, NumPy, SciPy, Matplotlib, scikit-learn
- Day 3, going deeper: tests, linting, pipelines, coding agents and reviewing their work





# More from Dataminded Academy



### Containerization with Docker

Pack a script with its Python and packages, so it runs the same anywhere

`github.com/datamindedacademy/containerization_with_docker`



### Orchestration with Airflow

Run scripts on a schedule, in order, with retries and alerts

`github.com/datamindedacademy/workflows_with_airflow`



### Polars and PySpark

Data frames for large logs: Polars on one machine, PySpark on a cluster

`github.com/datamindedacademy/dataframe-of-mind`




# Thank you

See you on day 2

